Fabrication and Integration Process

1. Material Choice Justification :

Initially, we had considered using **MDF** for the main build (no laser cutting, manual cutting and sanding), but since the holes, slits and cutouts throughout the model heavily depended on how we accommodate our circuit and GPIOs, we realized that **MDF** wouldn't be the preferable option (though it is extremely sturdy)

FINAL CHOICE:

We re-used **foamboards** that were lying around academic block 2.

They were:

- **Light**, extremely easy to work with.
- **Flexible**, easy to cut and shape, we were able to make last minute cutouts for the adaptor and laptop cable, which would have been difficult if it was MDF.
- **Sturdy**, when layered (the base).

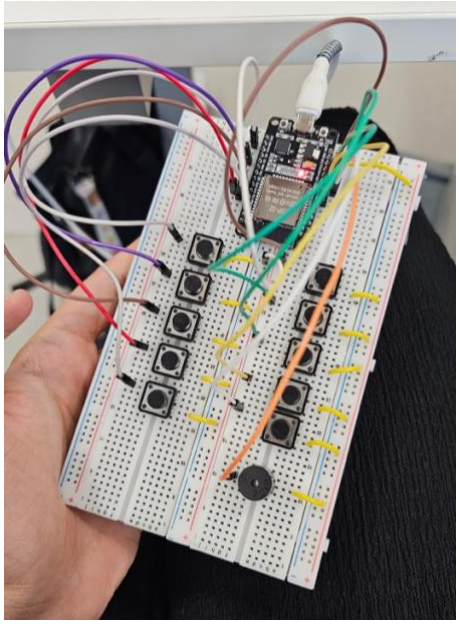
Thin cardboard was used for the door of the mini-locker (lightweight, less load and friction for rotation)

2. Fabrication Steps :

SOLDERING OF PUSH BUTTONS :D [I absolutely loved doing this]

For our first gateway, which involved OTP generation, we had to acquire a keypad on our locker to enter the generated OTP. The first thought that came into mind was to use push buttons :)

We required 10 pushbuttons for values 0-9. We tried the concept out on the bread board while prototyping and this is how it went:



The very first issue was that they **occupied most of the breadboard**. There were only a **limited number of arrangements** that we could go for and the amount of jumper wires getting tangled was just...outrageous.

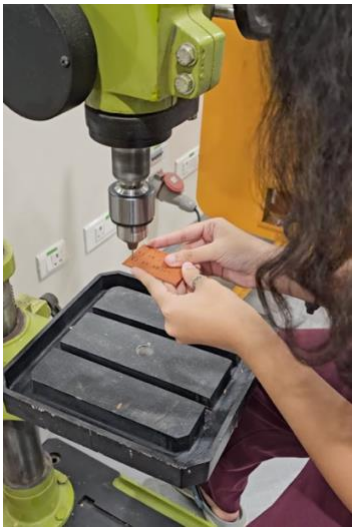
We also had to consider the fact that many more components were to be attached later on, and this setup wouldn't be preferable for the same.

On speaking with our faculty, Nayan, he recommended that we use a **"Vero" Board** for attaching all the pushbuttons.

Benefits of using a Vero Board:

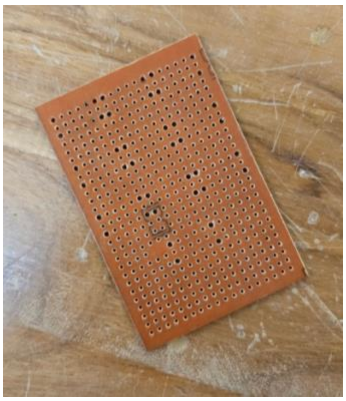
- Endless arrangement possibilities
- Each component can be connected individually without having to keep the series/parallel internal connections in mind, which was the case for the breadboard
- Easy to position around the final model.

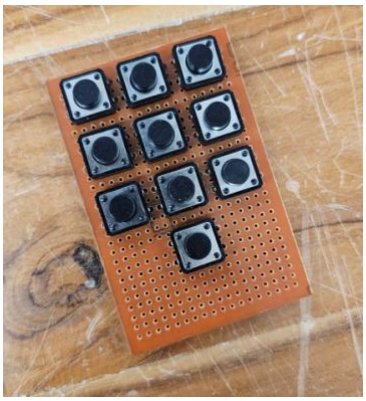
So, I managed to acquire two of them (extra, just in case I failed) from the **Digital Makerspace Lab.**



The holes on the **Vero Board** were small. The pins of the push buttons struggled to go through it. Hence, I had to resort to making the holes bigger, by using the **power drill machine** in the **Wood Workshop Lab.**

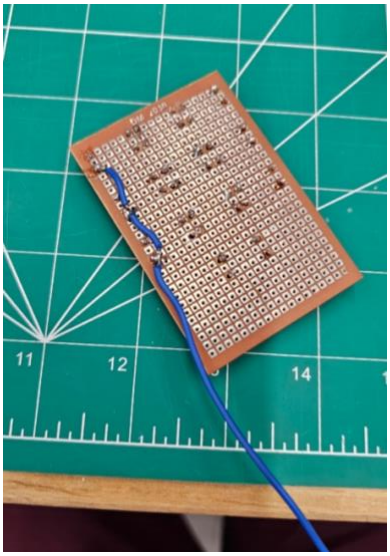
I had to mark the holes before drilling them.

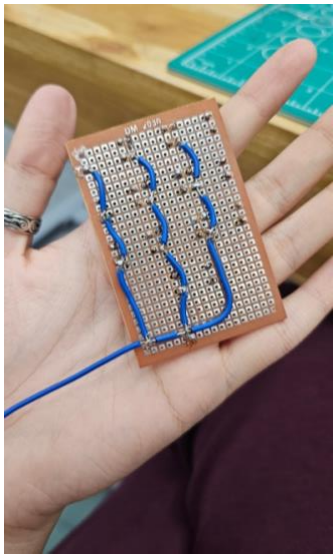




Bigger holes visible in the picture. Insertion of Pushbuttons made easier.

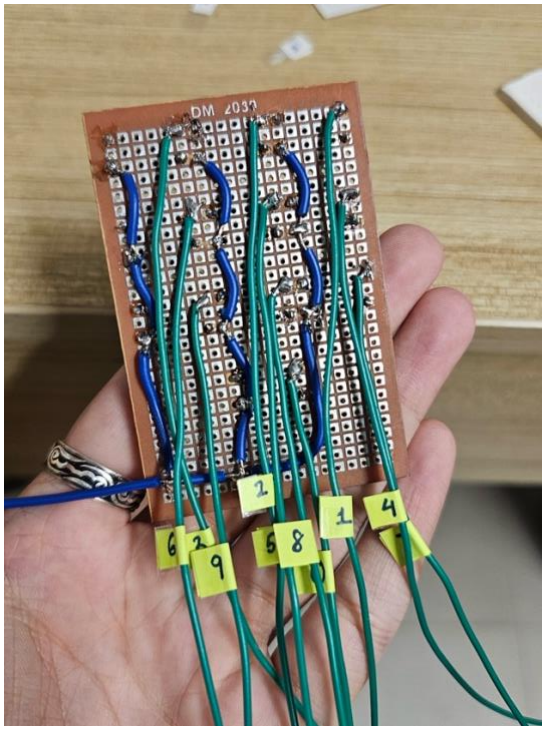
Arrangement imitates an actual keypad





Ideally, there would have been 10 wires for the GPIOs and 10 for GND. I wondered if there was way to combine **all the GNDs together** into 1 single wire.

On speaking with **Maharishi,** he said that I could use 3 individual wires, line them vertically, scrape of the insulation, and join them together into one at the end.



It was a lengthy process, but I had fun soldering it!

From the originally planned 20 wires, I brought it down to **11**.

I then **labelled** the keypad numbers on each wire for convenient connection on the GPIOs of the ESP32. (Thank God I did that)

IDEATION OF THE LOCKER

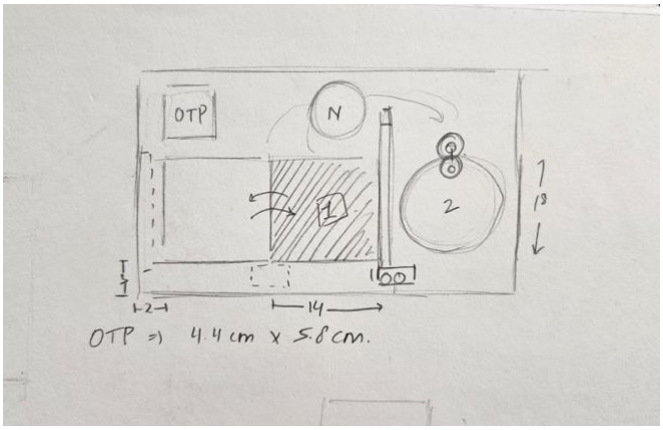
Dimensions : 45cm x 30cm x 30cm

Board count :
45cm x 30cm – **4** boards [Front, back, top, 2 for base to make it sturdy]
30cm x 30cm – **2** boards [Side walls]

We **painted** all the walls of the locker using **Black and White acrylic paint [to form grey].**

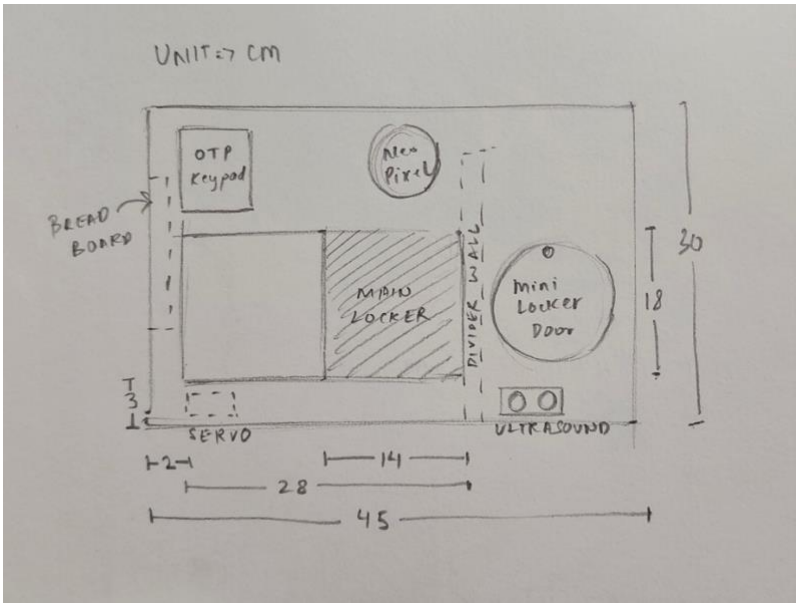
Ideation of placement of each component on the front:

First Iteration:



This was a rough iteration and dimensions of how each component would be placed.

I then cut newspaper pieces to confirm the dimensions.



FINAL FRONT VIEW.

Final iteration of how everything should be placed.

Points to note:

- A **divider wall** was to be placed between the main and mini locker, the height of which was decreased to accommodate the jumper wires that would pass through to the mini locker (to join the servo and HC-SR06).
- Placement of the **breadboard** on the left wall. This was done, to have it closest to the keypad (as most of the GPIOs extended from it).
- Elevation of the **main locker was done** (3cm from below) to accommodate the sliding mechanism wedge. (explained further)

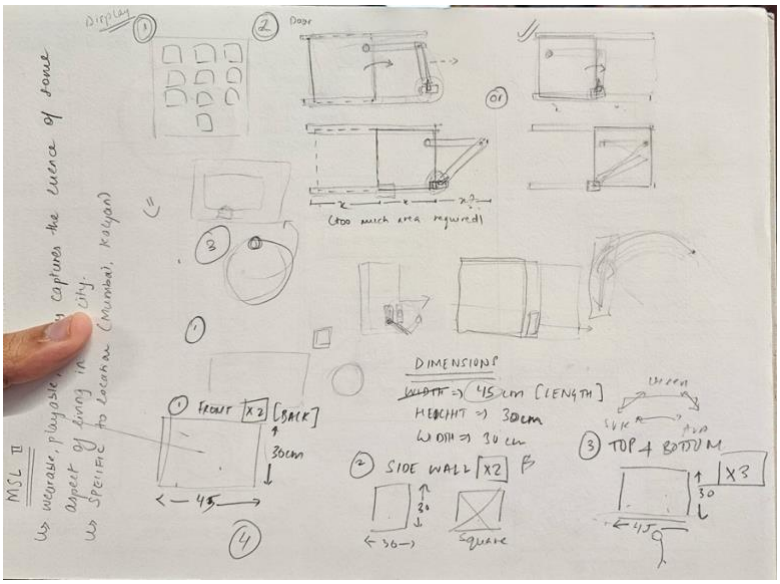
SLIDING MAIN DOOR [mechanism]

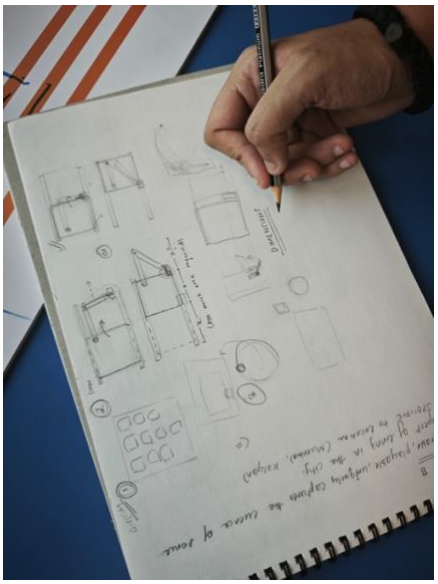
Originally, we were going to use the servo motor to create a normal hinge door mechanism, but on reviewing our concept with Nayan, he recommended that we explore more methods of making our model **appealing**.

We decided to execute a sliding door mechanism.

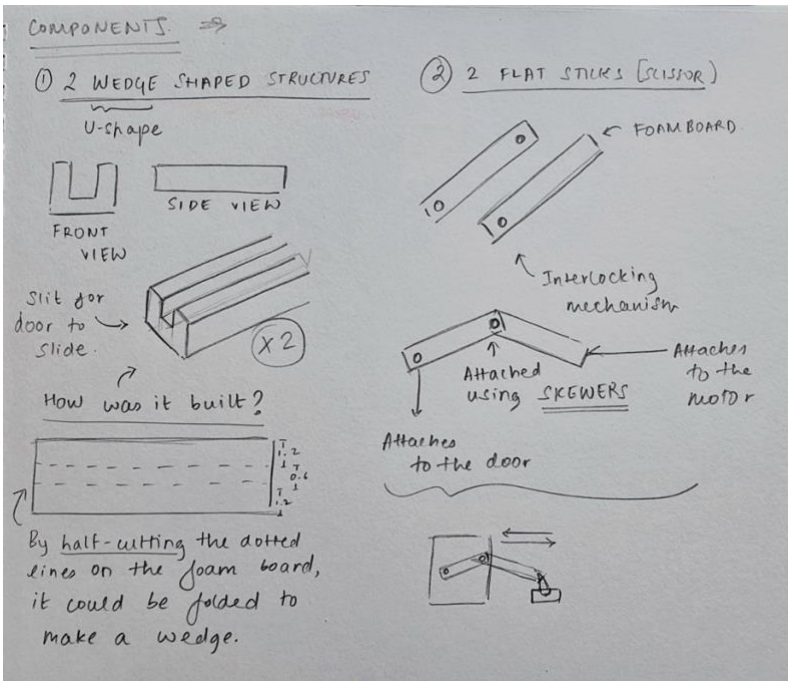
It involved having **2 stick-like components**, to create a flexible **scissor** structure that would expand and contract when the servo moves, making the door open and close.

Rough ideation of the door mechanism:

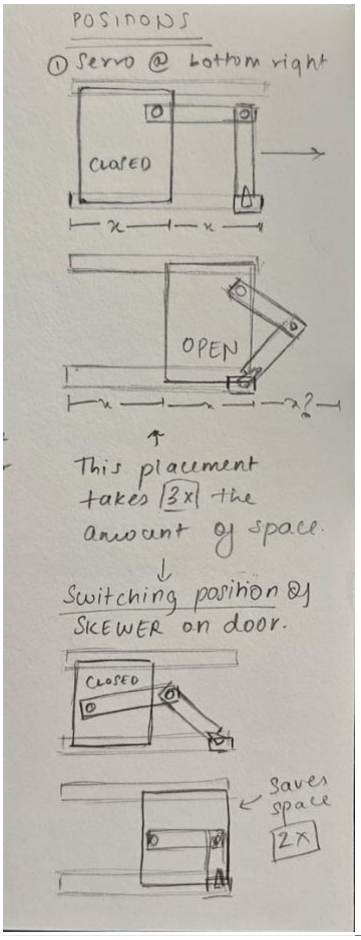


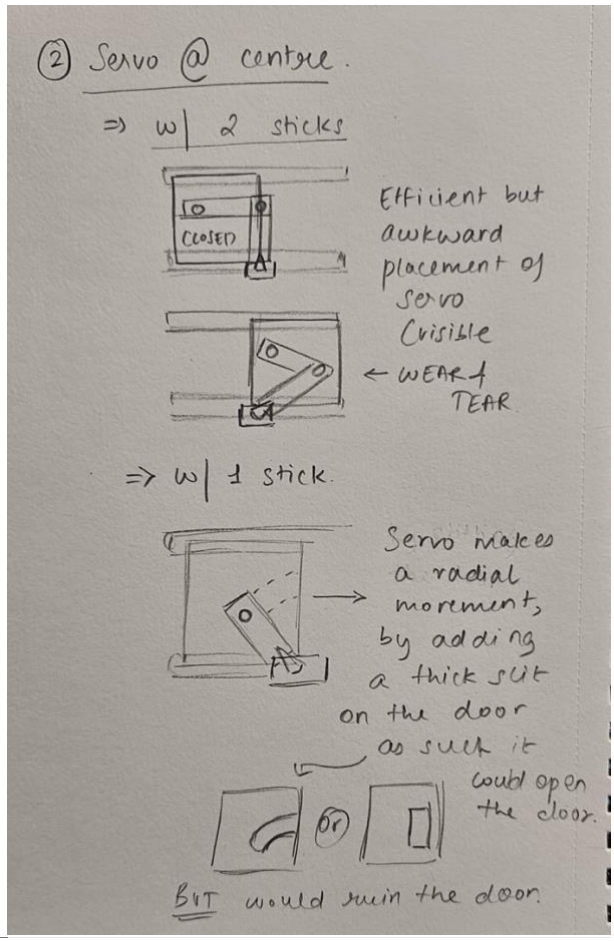


Components:

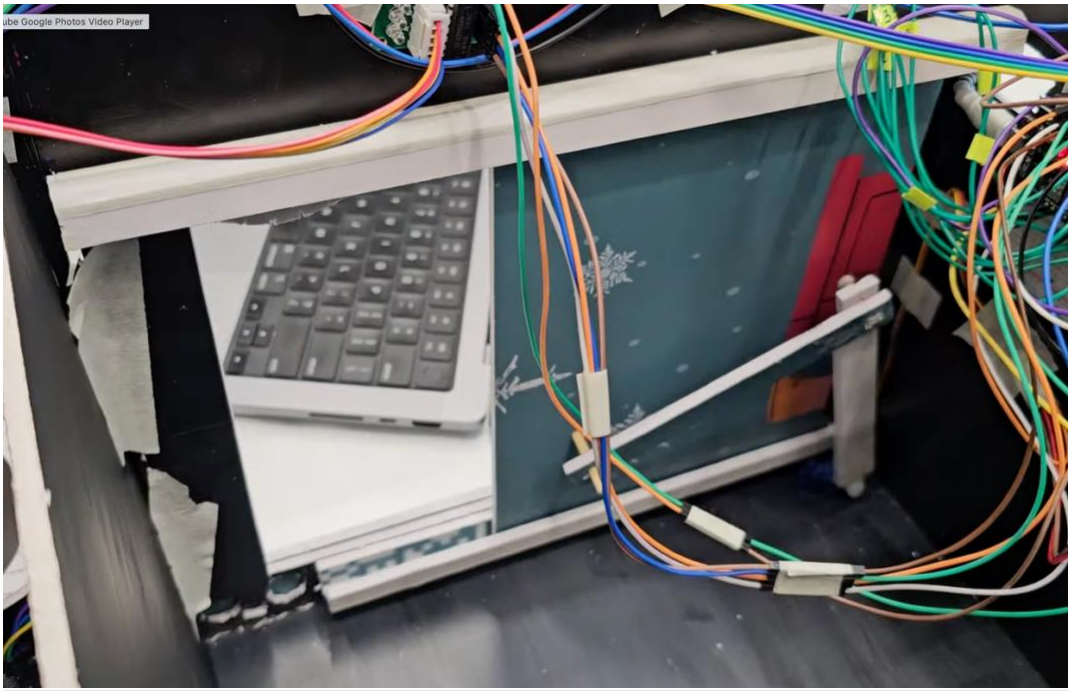


Explanation:





Position 1 was finalized, with the servo at the right.



***Final Build** on the main locker

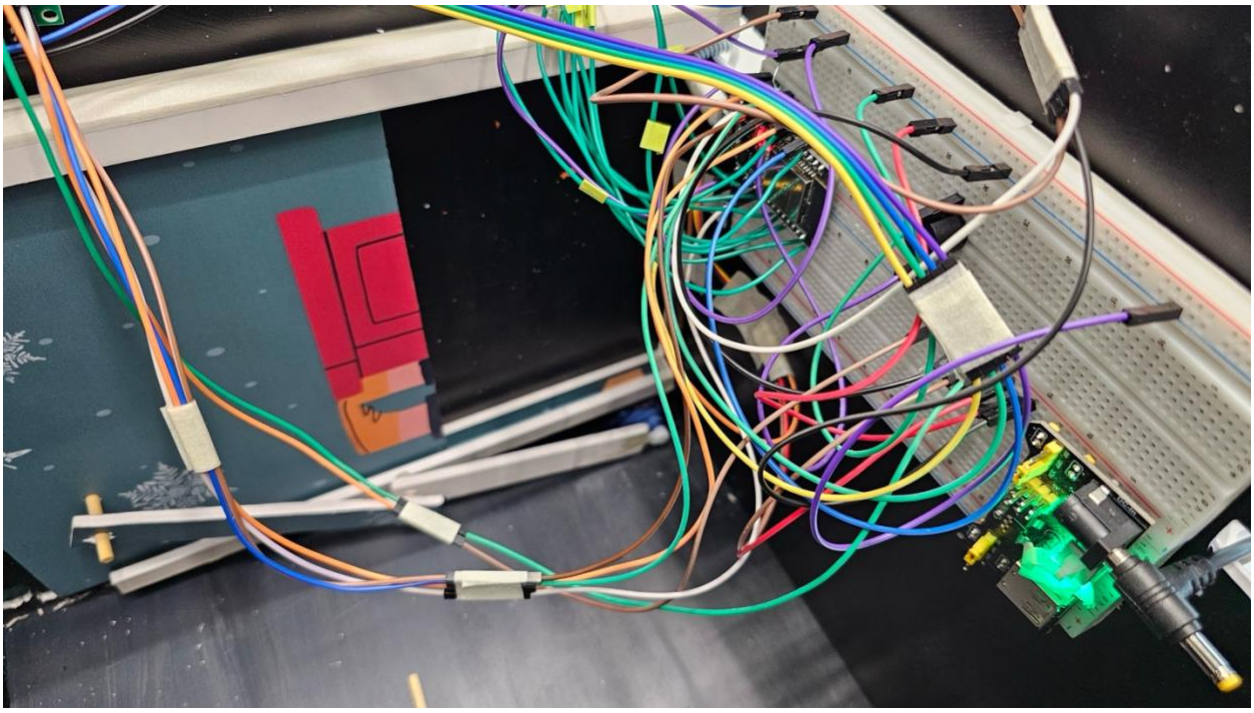
FINAL ASSEMBLY OF LOCKER

The foam boards were then joined together using fevibond and a lot of patience.

The 2 wedges of the door mechanism were fixated on the front wall.

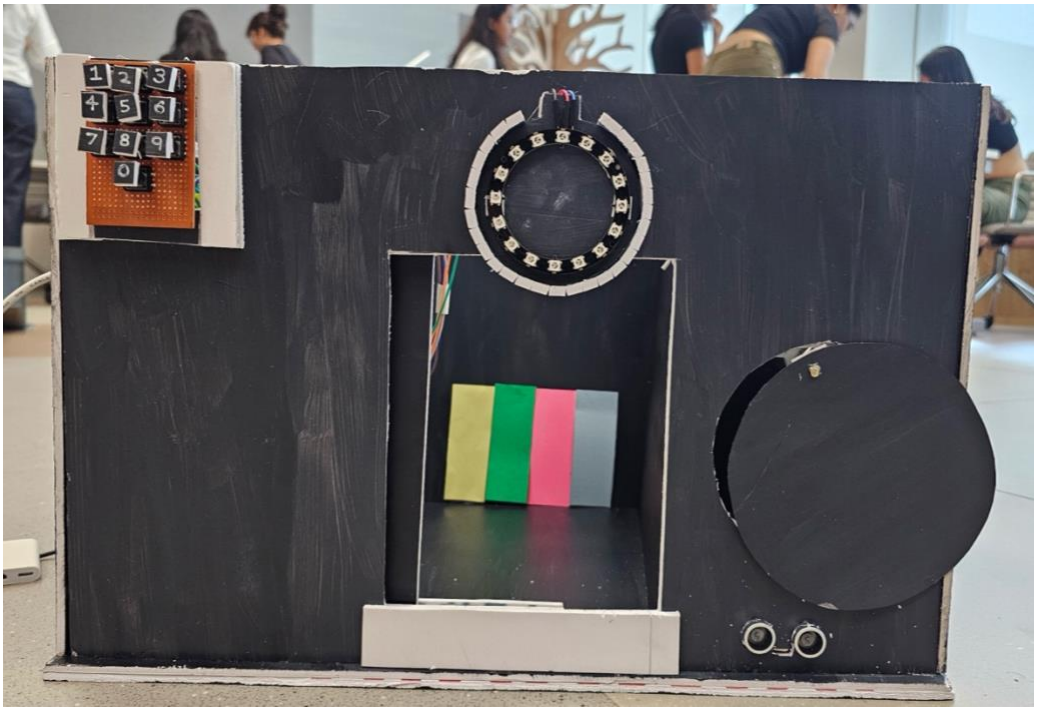
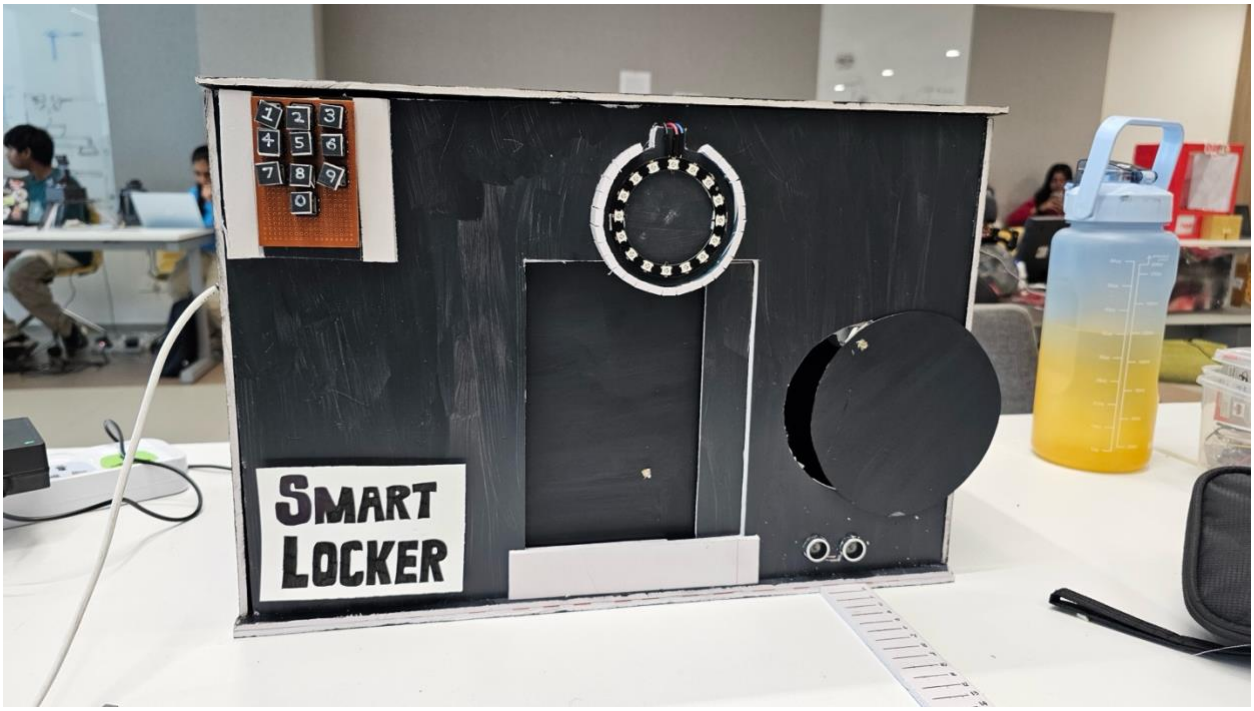
We did **not fix** the top board for easy access to the circuit

I assembled the final circuit within the structure (made sure to tape everything too)



Final **aesthetic touches** (using foamboard) were added to make the locker more appealing.

- Number pieces on the **keypad**
- **Elevation** of keypad using foamboard for users to easily enter the OTP.
- A structure for the **NeoPixel** to elevate it ahead.



Coloured sheets were added within the main locker for the mini locker unlocking sequence.